

FYP PROJECT PROPOSAL

Project Title

E-commerce Product Demand Predictor



Groups Members :

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Supervisor

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1. Introduction

E-commerce businesses face challenges in managing inventory and forecasting demand for products (SKUs). Overstocking or understocking leads to financial losses and missed opportunities. This project proposes an AI-powered demand prediction system that analyzes historical sales and web traffic data to forecast product demand, identify slow-moving items, and suggest marketing strategies to optimize sales.

2. Problem Statement

- Online stores often lack accurate demand forecasts.
- Manual prediction methods are time-consuming and error-prone.
- Businesses cannot easily identify slow-moving products for targeted marketing.
- Goal: Develop a system that provides reliable demand forecasts and actionable insights for e-commerce businesses.

3. Objectives

- Predict future demand for products using historical sales and web traffic data.
- Recommend marketing strategies for low-demand products.
- Integrate with Daraz for real-time data updates.
- Provide a dashboard to visualize trends, predictions, and insights.

4. Scope

- Predict demand for e-commerce products on a weekly or monthly basis.
- Provide actionable recommendations for slow-moving products.
- Visualize trends and predictions on a simple dashboard.
- Optional API integration for live sales data.

5. Functionalities

- Upload historical sales and web traffic data (CSV/Excel).
- Predict demand for individual products (SKU-level).
- Identify slow-moving and high-demand products.
- Generate marketing recommendations (discounts, promotions, bundling).
- Dashboard to visualize predictions, trends, and insights.
- Optional: Fetch live sales data from Daraz.

6. Key Modules

- Module Name Description
- Data Input Module Upload or fetch sales/traffic data (CSV/API integration)
- Data Preprocessing Module Clean, merge, and prepare data for analysis
- Prediction Module Use ML models (Linear Regression, Random Forest, LSTM) to forecast demand
- Recommendation Module Identify slow-moving products and suggest marketing strategies
- Dashboard/Visualization Display predictions, trends, and product insights using charts and graphs
- Optional API Integration Connect to Daraz for live data updates

7. Comparison Table

Feature / System	Proposed System	System A (Shopify Analytics)	System B (Zoho Inventory)	System C (Google Analytics)
SKU-level demand prediction	Yes	Limited	Limited	Limited
AI/ML-based forecasting	Yes	No	No	No
Marketing recommendations	Yes	No	No	No
Real-time API integration	Yes	daraz	No	No
Dashboard Visualizations	Interactive (Plotly/Charts)	Basic charts	Basic charts	Website traffic only
Focus on slow-moving products	Yes	Limited	Limited	No

Explanation: The proposed system is unique because it combines AI-driven demand forecasting with marketing insights, something most existing platforms do not provide.

8. Methodology

- Data Collection: Collect historical sales data and web traffic data.
- Data Preprocessing: Clean and merge datasets, create features.
- Prediction Model: Train ML models (Random Forest, LSTM) to forecast product demand.

- Recommendation Engine: Identify slow-moving products and generate strategies.
- Dashboard Development: Build a web-based interface for visualization.
- Evaluation: Measure accuracy using RMSE/MAE and compare predicted vs actual demand.

9. Tech Stack

- Layer Technology / Tool
- Backend Python, Flask / Django
- Database MySQL / PostgreSQL
- Machine Learning Scikit-learn, TensorFlow/Keras, XGBoost
- Visualization Plotly, Matplotlib, Dash
- API Integration Daraz REST API
- Frontend HTML, CSS, Bootstrap
- 10. Expected Output / Deliverables
- Accurate demand predictions for e-commerce products.
- Dashboard showing product trends and predictions.
- Recommendations for low-demand products.
- Optional: Integration with live e-commerce data via API.